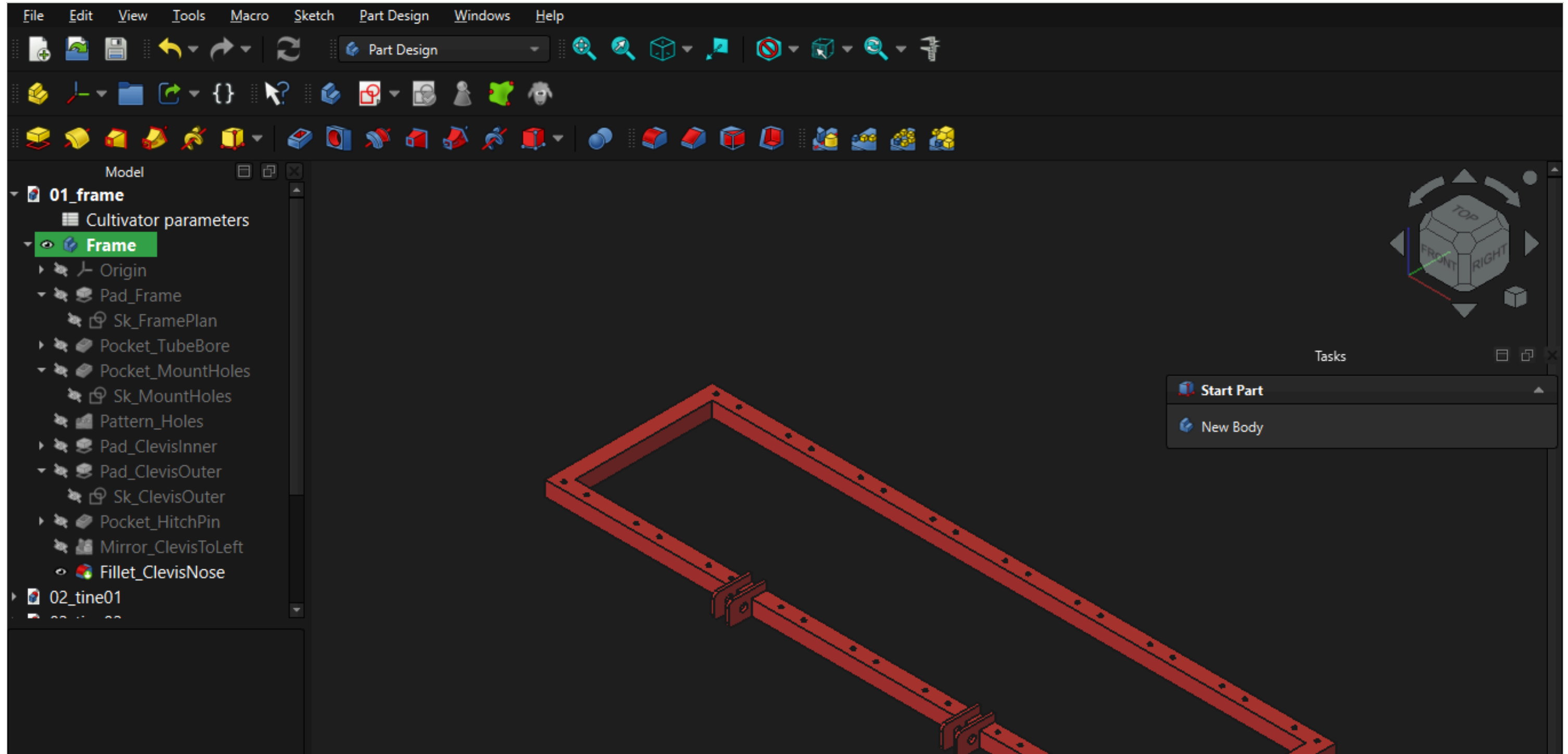
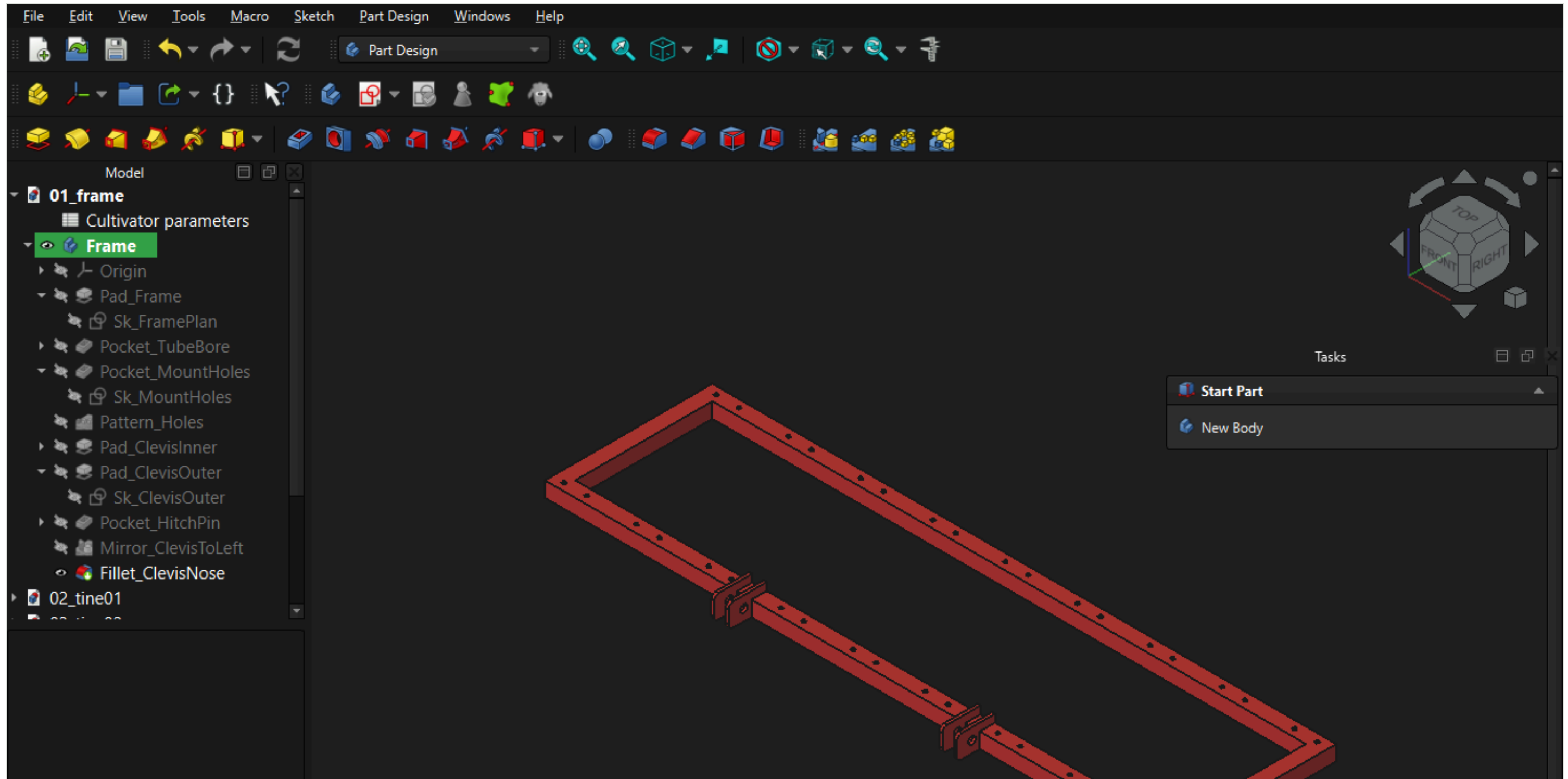


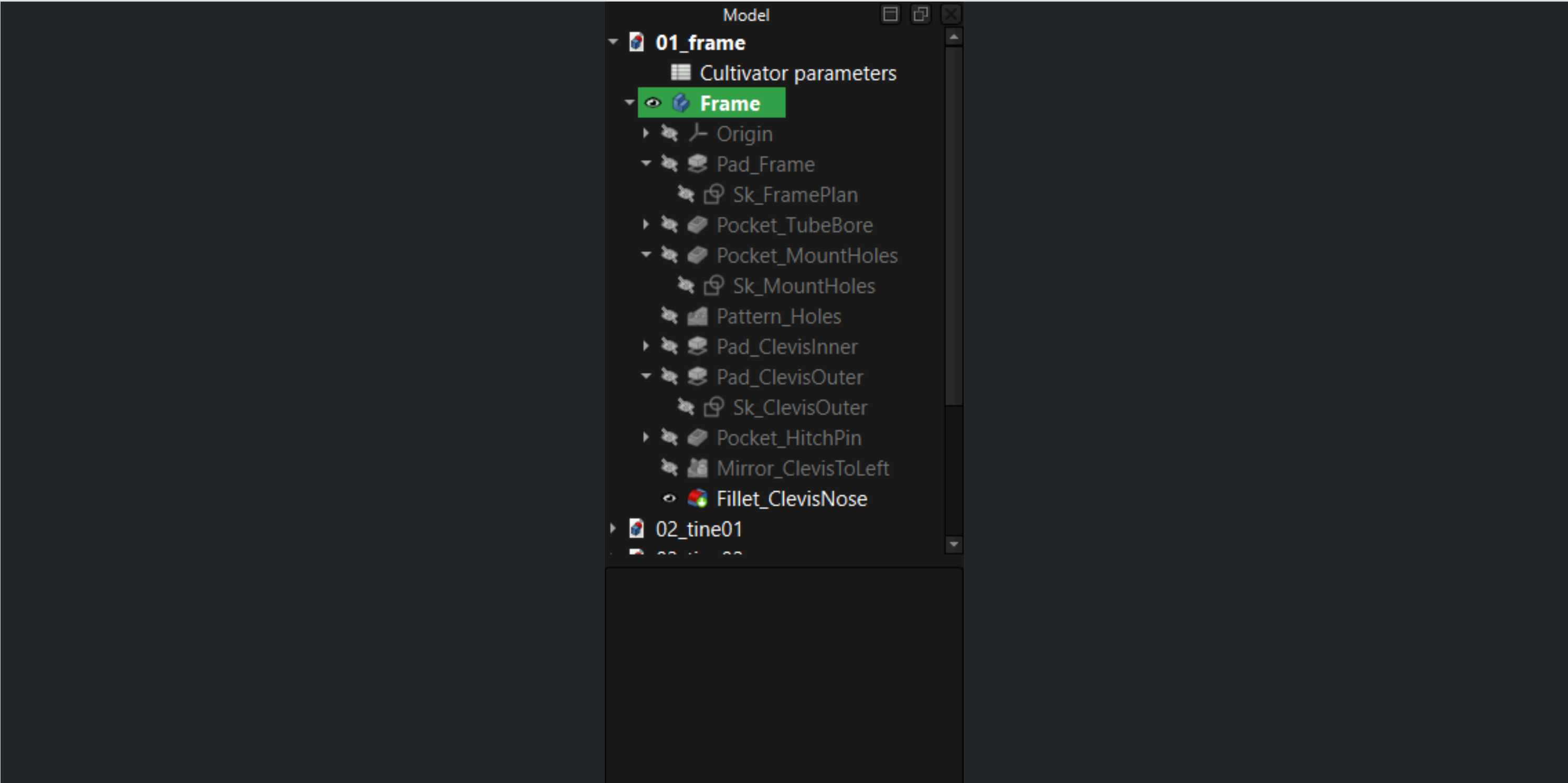
The FreeCAD Window



Five parts of the window

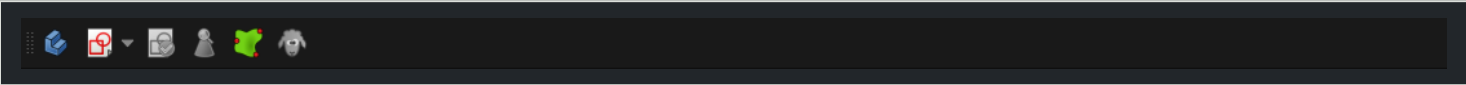


The tree is your build history

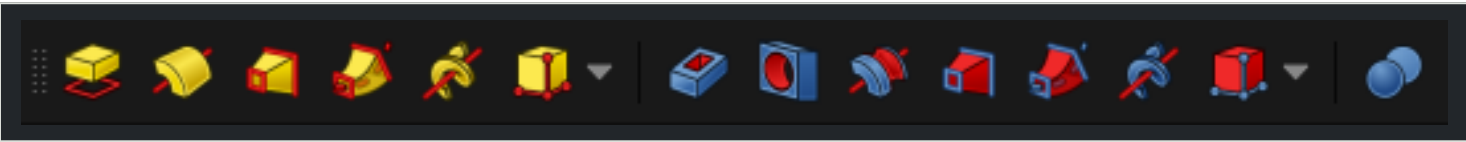


Part Design has four toolbars

Learn what each group is for and you will stop hunting for icons.



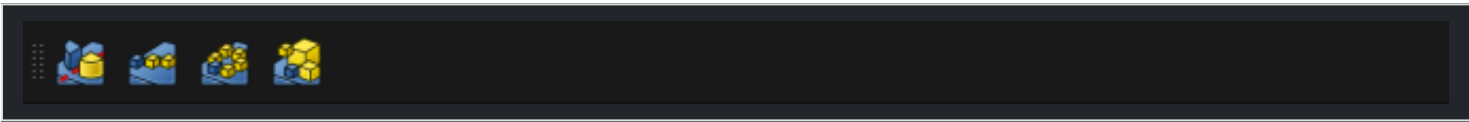
Helper. Make a body, make a sketch, make a datum plane.



Modelling. Everything that adds or removes metal.



Dress-up. Fillet, chamfer, draft, thickness.



Transformation. Mirror and pattern.

Adding metal and removing metal

Every modelling command either adds material or takes it away. They work in pairs.

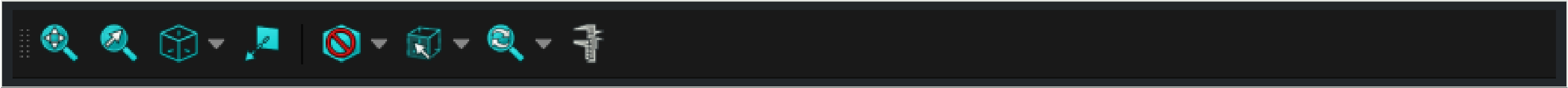
ADDS	REMOVES	MADE FROM
Pad	Pocket	a sketch, pushed straight
Revolution	Groove	a sketch, spun about an axis
Additive Loft	Subtractive Loft	two or more sketches
Additive Pipe	Subtractive Pipe	a sketch pushed along a path

DRESS-UP IS DIFFERENT

- Fillet, Chamfer and Draft do not use a sketch.
- They work on the faces and edges of the solid you already have.
- That makes them fragile. Add them last, and use as few as you can.

On the frame we ask for a fillet that is physically impossible, and FreeCAD refuses. Deck 5 walks through it.

Getting around the 3D view



The View toolbar: fit, draw style, standard views

Six number keys and one letter will cover almost everything you need.

KEY	WHAT IT DOES
1 to 6	Front, top, right, rear, bottom, left
0	Isometric view
V then F	Fit the model on screen
Space	Show or hide whatever is selected
Scroll wheel	Zoom
Middle button drag	Pan

IF YOU ARE WATCHING THE SOLIDWORKS VIDEO TOO

SolidWorks words, FreeCAD words

SOLIDWORKS	FREECAD
Units: IPS	Imperial decimal
FeatureManager tree	Model tree
Front / Top / Right plane	XZ / XY / YZ plane
Smart Dimension	Dimension constraint
Fully Defined	Fully constrained
Mirror Entities	Symmetry constraint
Boss-Extrude	Pad
Extruded Cut	Pocket

SOLIDWORKS	FREECAD
Blind	Type: Dimension
Through All	Type: Through all
Mid Plane	Symmetric to plane
Linear Pattern	LinearPattern
Mirror	Mirrored
Hole Wizard	Hole
Mates	Joints

Careful with the planes. SolidWorks calls it the **Right** plane; FreeCAD calls the same plane **YZ**. Most of the tyne and clamp sketches in this course go on it.

Three things SolidWorks has that FreeCAD does not

You will meet all three in this machine. In two cases the FreeCAD way is actually better.

SOLIDWORKS TOOL	WHAT WE DO INSTEAD	DECK
Weldments — structural members	Sketch the tube section and Pad it, or sweep it along a path with Additive Pipe	5, 8
Flex — bending a finished plate	Build the blade curved from the start, by sweeping it along an arc of the radius we want	9
Linear Component Pattern in an assembly	Use a link array, which is one object holding a list of positions	10, 11

Because the blade is built curved, its bend radius is just a number in a spreadsheet. Blade 02 is Blade 01 with that number changed from 10 to 16.